

How do mRNA vaccines work?

SUMMARY — The shot delivers a fatty droplet carrying genetic instructions. Your own cells read them, build one harmless viral protein, and the immune system learns to attack it. Two chemical tricks made it practical: modified letters that stop the body destroying the message, and the droplet, which both smuggles it into cells and acts as a built-in immune stimulant. Protection against severe disease is strong and lasting; against infection it is neither.

Unedited example output. Size: medium, mode: image.
Prompt: "Use the scientific-review skill, medium image mode: how do mRNA vaccines work?"

The vaccine carries a message, not a pathogen

- It is a lipid nanoparticle (LNP) wrapping messenger RNA (mRNA); the particle enters cells, releases its cargo from the endosome, and ribosomes translate it into protein Chaudhary et al. 2021, Pardi & Krammer 2024.
- Both licensed COVID-19 vaccines encode the same antigen — a prefusion-stabilised SARS-CoV-2 spike protein, a design ported from earlier coronavirus structural work — at 30 μg (BNT162b2) or 100 μg (mRNA-1273) Polack et al. 2020, Baden et al. 2021, Corbett et al. 2020.
- The mRNA is transient and non-integrating, so a new vaccine means a new sequence, not a new process Pardi & Krammer 2024; yield depends on the untranslated regions, codon usage and folding Jin et al. 2025.

Chemically modified RNA letters were the unlock

- Modified nucleosides such as pseudouridine abolish RNA's ability to trigger Toll-like receptors 3, 7 and 8, so exposed dendritic cells make far less cytokine Karikó et al. 2005.
- Stripping double-stranded RNA contaminants by chromatography raised translation a further **10- to 1,000-fold** in primary cells Karikó et al. 2011.

- N1-methylpseudouridine (m1 Ψ), used in both licensed vaccines, beat pseudouridine by up to ~ 13 -fold in mice Andries et al. 2015; it raises output by altering translation dynamics, not only by damping innate sensing Rozman et al. 2026.
- The chemistry has a cost: m1 Ψ causes +1 ribosomal frameshifting, and immunity to the off-target products was detected in mice and humans — no adverse outcomes reported, and fixable by recoding Mulrone et al. 2024.

The lipid nanoparticle is a delivery vehicle and an adjuvant

- LNPs are ionizable-lipid carriers: charged at low pH to load RNA, near-neutral in blood, which limits toxicity Cullis & Felgner 2024, Hou et al. 2021.
- In mice the formulation is itself an adjuvant, outperforming an MF59-like comparator and driving T follicular helper (Tfh) and antibody responses through the ionizable lipid and interleukin-6 Alameh et al. 2021.
- The parts divide the labour: the mRNA drives type I interferon that matures dendritic cells, while the LNP shapes where those cells sit in the draining lymph node Castaño et al. 2025.

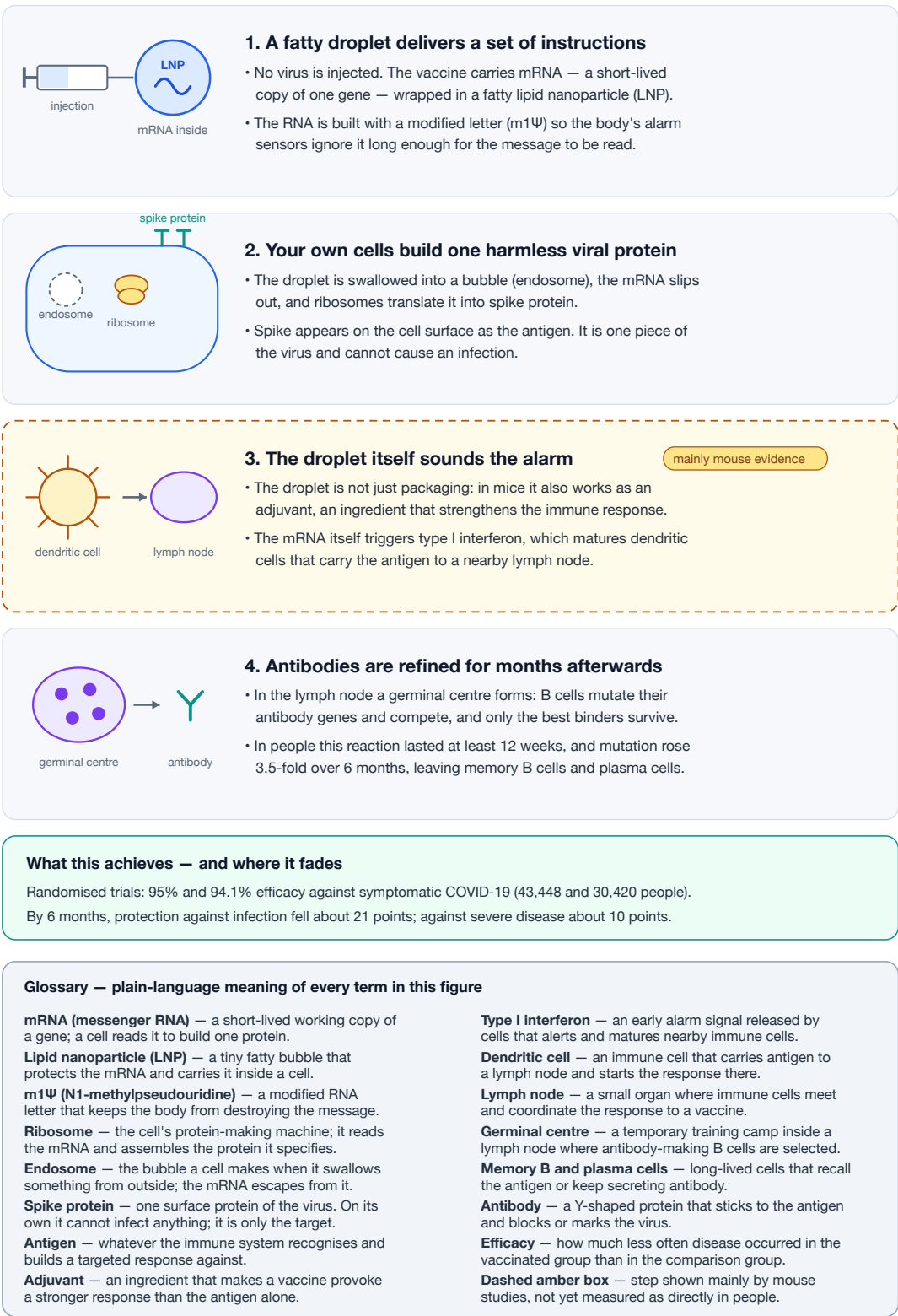
T cells are primed by an unconventional route, and antibodies are refined for months

- In the first human trial, BNT162b1 elicited TH1-skewed CD4 and CD8 T-cell responses plus above-convalescent antibodies (phase 1/2) Sahin et al. 2020.

- Priming skips the textbook route: in mice it needs neither cDC1 dendritic cells nor classical cross-presentation, drawing on "cross-dressing" with ready-made peptide–MHC class I complexes Jo et al. 2026, and gave about ten times more antigen-specific CD8 cells than protein plus an immunostimulant Muro et al. 2026.
- Unlike an adjuvanted protein vaccine, mRNA generated germinal centre B and Tfh cells in mice, with germinal-centre size tracking neutralising antibody levels Lederer et al. 2020.
- In people, lymph-node sampling (n=14) found germinal-centre B cells for at least **12 weeks**; across 1,540 clones somatic hypermutation rose **3.5-fold over six months**, leaving higher-affinity memory B cells, bone-marrow plasma cells and Tfh cells lasting about six months Turner et al. 2021, Kim et al. 2022, Mudd et al. 2022.

How an mRNA vaccine works

From an injected fatty droplet to antibodies that keep improving for months



the route from injection to durable antibody — droplet uptake, translation of modified-nucleoside mRNA into spike protein, and a germinal-centre reaction that keeps improving antibodies for months [Pardi & Krammer 2024](#), [Turner et al. 2021](#), [Kim et al. 2022](#). The dashed amber panel marks the step resting mainly on mouse experiments [Alameh et al. 2021](#), [Castaño et al. 2025](#); the other steps are observed in vaccinated people. Schematic, not to scale.

Head to head, the RNA platform outperforms the alternatives

- A meta-regression of 35 randomised trials put symptomatic COVID-19 **70% lower** on mRNA than on viral-vector vaccines (incidence rate ratio 0.30, 95% CI 0.19–0.46) and 37% lower on protein-subunit vaccines (0.63, 0.46–0.86); inactivated vaccines showed no advantage (0.98, 0.71–1.36) Goryaynov & Gurova 2024.

- Randomised mixing confirms it within-trial: an mRNA boost after a vector prime multiplied anti-spike IgG **nine- to tenfold** over a second vector dose (n=463; n=1,072); the reverse order missed non-inferiority Liu et al. 2021, Stuart et al. 2022.
- In a network meta-analysis of 63 studies, three mRNA doses ranked best against hospital admission (96%, 95% credible interval 82–99 pre-Omicron; 93%, 76–98 after) Au & Cheung 2022.
- A self-amplifying RNA vaccine also beat a vector vaccine head-to-head: relative efficacy 19.8% (4.0–33.0) over a year (n=2,366) Ho et al. 2024.

VACCINE (DOSE)	DESIGN	N	EFFICACY	REF
BNT162b2 mRNA (30 µg ×2)	RCT vs placebo	43,448	95% (95% CrI 90.3–97.6) vs symptomatic COVID-19	<u>Polack et al. 2020</u>
mRNA-1273 mRNA (100 µg ×2)	RCT vs placebo	30,420	94.1% (89.3–96.8) vs symptomatic COVID-19	<u>Baden et al. 2021</u>
ARCT-154 self-amplifying (5 µg ×2)	RCT vs placebo, Delta era	16,100	56.6% (48.7–63.3) any; 95.3% (80.5–98.9) severe	<u>Hfö et al. 2024</u>
ARCT-154 vs ChAdOx1-S	RCT, head-to-head	2,366	19.8% (4.0–33.0) relative efficacy	<u>Ho et al. 2024</u>
mRNA-1010 influenza	RCT vs licensed standard dose, ≥50 y	40,703	26.6% (16.7–35.4) relative efficacy	<u>Leroux-Roels et al. 2026</u>

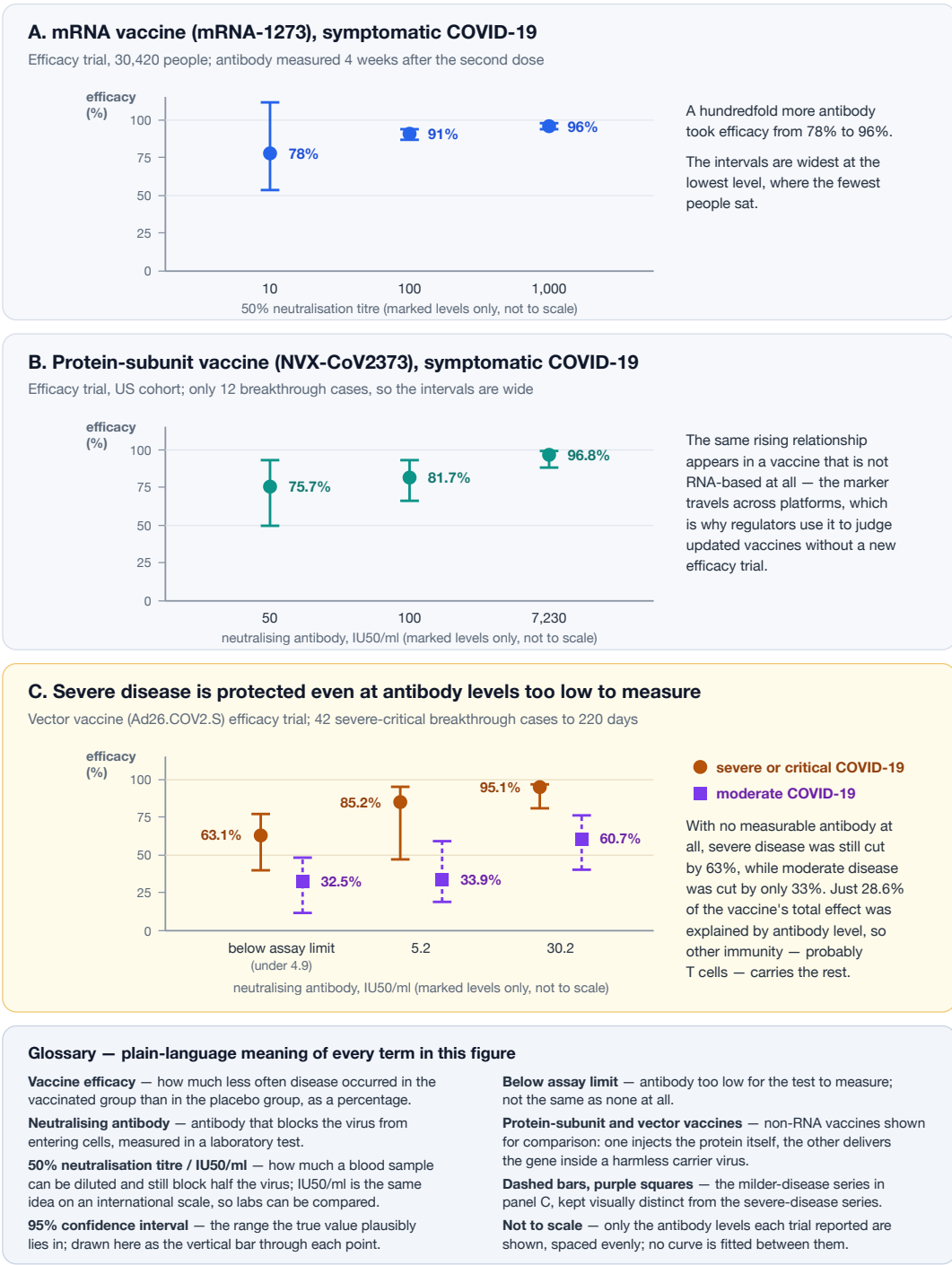
Antibody level tracks protection, but does not account for all of it

- In the mRNA-1273 trial, vaccine efficacy rose with neutralising titre: 78% (95% CI 54–89) at a titre of 10, 91% (87–94) at 100, 96% (94–98) at 1,000 Gilbert et al. 2022; it held for a third dose in the Omicron era (hazard ratio 0.31, 0.10–0.96 per tenfold rise) Zhang et al. 2024.
- The same gradient appears for a protein-subunit vaccine Fong et al. 2023 and for the RSV vaccine mRNA-1345 (hazard ratio 0.44, 0.30–0.65) Ma et al. 2025; pseudovirus neutralisation is the strongest single marker Benkeser et al. 2023.

- Antibody is not the whole mechanism: in a vector-vaccine trial with 42 severe breakthroughs, efficacy against severe disease was **63.1%** (40.0–77.3) even below the assay's limit, and only **28.6%** (8.5–48.7) of total efficacy was mediated by titre Carpp et al. 2024.
- T-cell receptor sequencing in 396 trial participants linked vaccine-elicited T-cell responses to lower risk of **severe** COVID-19 but not of symptomatic COVID-19 overall Hertoghs et al. 2025; where prior infection is common, titre alone is a poor guide Hwang et al. 2025.

More antibody, more protection — but antibody is not the whole story

Vaccine efficacy at the antibody levels each trial reported. Bars show 95% confidence intervals.



vaccine efficacy at the antibody levels each trial actually reported — real estimates with their intervals, evenly spaced, with no curve fitted between them. Efficacy rises with antibody in the mRNA-1273 trial [Gilbert et al. 2022](#) and in a protein-subunit trial [Fong et al. 2023](#), which is why the marker is used as a cross-platform surrogate [Benkeser et al. 2023](#). Panel C is the caveat: severe disease was still cut by 63% at unmeasurable antibody levels, and only 28.6% of total efficacy was mediated by antibody [Carpp et al. 2024](#) — consistent with T-cell responses tracking protection from severe but not from mild disease [Hertoghs et al. 2025](#).

Protection wanes fastest against infection, and re-sequencing only partly fixes it

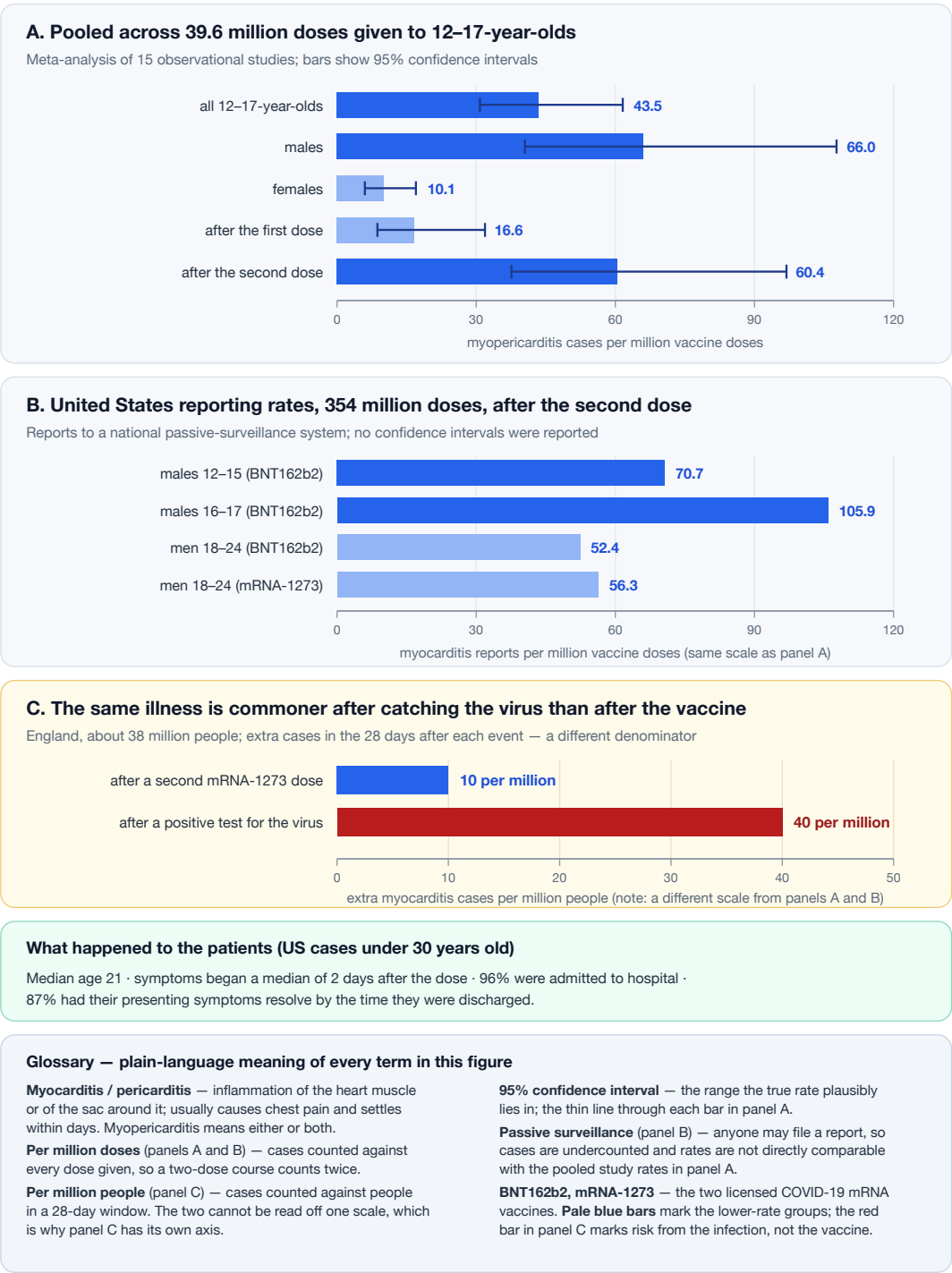
- Across 78 estimates, effectiveness fell **21.0 percentage points** (95% CI 13.9–29.8) against infection between 1 and 6 months, but only **10.0** (6.1–15.4) against severe disease [Feikin et al. 2022](#).
- One reason is anatomical: bronchoalveolar sampling found little neutralising antibody and no spike-specific B or T cells in vaccinated airways, unlike convalescents, despite strong blood responses [Tang et al. 2022](#).
- Updating the sequence chases a moving target: a bivalent Omicron booster raised BA.1 titres 1.6-fold over the ancestral one (n=814) [Chalkias et al. 2022](#), yet BA.5-bivalent sera barely neutralised the BQ.1.1 and XBB.1 variants that followed [Kurahade et al. 2023](#), and pooling 12 studies (4,581 people) showed no gain over variant-matched monovalent boosters [Banga Ndzouboukou et al. 2025](#).
- They still work where it matters: across 1.88 million matched pairs aged ≥65, an XBB.1.5 dose gave 57.9% (49.9–65.8) effectiveness against hospital admission and 75.2% (70.6–79.9) against death over 24 weeks [Andersson et al. 2024](#).
- Repeated dosing leaves marks: responses stay imprinted toward the strain first seen (original antigenic sin) [Song et al. 2024](#), [Baerends et al. 2023](#), and boosters raise spike-specific IgG4, which correlated inversely with non-neutralising antibody functions [Selva et al. 2026](#).

Common side effects are mild; myocarditis is the one real rare harm

- Active surveillance of 11.8 million doses found no serious outcome above the pre-specified signal threshold, and anaphylaxis at **4.8** (3.2–6.9) and **5.1** (3.3–7.6) per million doses of BNT162b2 and mRNA-1273, by an unresolved mechanism [Klein et al. 2021](#), [Risma et al. 2021](#).
- The exception is myocarditis and pericarditis, confirmed as mRNA-specific across 99 million vaccinated people; the same study put Guillain-Barré syndrome and cerebral venous sinus thrombosis after the vector vaccine, not after mRNA [Faksova et al. 2024](#).
- It is rare and concentrated: 39.6 million adolescent doses gave **43.5** cases per million (30.8–61.6), 6.5 times higher in males, 3.6 times higher after the second dose, and lower than after infection at the same ages [Guo et al. 2023](#); US surveillance peaked at 105.9 per million second doses in males aged 16–17 [Oster et al. 2022](#), and in England the excess after a second mRNA-1273 dose was 10 per million against 40 after a positive test [Patone et al. 2022](#).
- Against that, in 45.7 million adults arterial and venous thrombosis were **lower** after every dose and brand, myocarditis rising only transiently (adjusted hazard ratio 3.14, 2.04–4.85 in the week after a second BNT162b2 dose) [Ip et al. 2024](#).

How rare is myocarditis after mRNA vaccination?

Cases per million, from three independent datasets. The three panels use two different denominators — see the glossary.



three independent datasets on the one confirmed mRNA-specific harm. Panel A pools 15 studies covering 39.6 million adolescent doses [Guo et al. 2023](#); panel B is passive surveillance, which undercounts and is therefore not directly comparable — it also supplies the clinical course shown in the strip [Oster et al. 2022](#); panel C uses a different denominator, extra cases per million people over 28 days, and puts the risk after infection above the risk after a second mRNA-1273 dose [Patone et al. 2022](#). The signal is real at population scale [Faksova et al. 2024](#), while overall thrombotic events were lower after vaccination than without it [Ip et al. 2024](#).

The binding constraints are chemical and logistical, and the next generation targets both

- Manufacturing and cold chain, not immunology, are the practical limit [Rosa et al. 2021](#), [Pardi & Krammer 2024](#); the instability is intrinsic — backbone hydrolysis, nucleobase oxidation and freeze–thaw aggregation [Algaissi & Moni 2026](#).
- Freeze-drying works but is formulation-dependent: mRNA-LNPs survived lyophilisation in 20% sucrose, yet cargo retention and translation depended on the ionizable lipid [Lamoot et al. 2023](#), [Haensler et al. 2025](#).
- Self-amplifying RNA carries its own replicase, so far less is needed: ARCT-154 reached 56.6% efficacy against any COVID-19 and 95.3% against severe disease at 5 µg, one-sixth of the BNT162b2 dose [Hô et al. 2024](#), [Blakney et al. 2021](#).
- Beyond COVID-19: against a licensed standard-dose influenza vaccine in 40,703 adults aged ≥50, mRNA-1010 cut confirmed influenza-like illness by a further 26.6% (16.7–35.4), at the cost of more reactogenicity (injection-site pain 65.8% vs 29.8%) [Leroux-Roels et al. 2026](#); trials against other respiratory viruses are under way [Whitaker et al. 2023](#).

REFERENCES

- [Alameh et al. 2021](#) Lipid nanoparticles enhance the efficacy of mRNA and protein subunit vaccines by inducing robust T follicular helper cell and humoral responses. *Immunity*. <https://doi.org/10.1016/j.immuni.2021.11.001>
- [Algaissi & Moni 2026](#) Next-generation mRNA vaccines: strategies to overcome stability challenges. *npj Vaccines*. <https://doi.org/10.1038/s41541-026-01502-z>
- [Andersson et al. 2024](#) Comparative effectiveness of monovalent XBB.1.5 containing covid-19 mRNA vaccines in Denmark, Finland, and Sweden: target trial emulation based on registry data. *BMJ Medicine*. <https://doi.org/10.1136/bmjmed-2024-001074>
- [Andries et al. 2015](#) N1-methylpseudouridine-incorporated mRNA outperforms pseudouridine-incorporated mRNA by providing enhanced protein expression and reduced immunogenicity in mammalian cell lines and mice. *Journal of Controlled Release*. <https://doi.org/10.1016/j.jconrel.2015.08.051>
- [Au & Cheung 2022](#) Effectiveness of heterologous and homologous covid-19 vaccine regimens: living systematic review with network meta-analysis. *BMJ*. <https://doi.org/10.1136/bmj-2022-069989>
- [Baden et al. 2021](#) Efficacy and Safety of the mRNA-1273 SARS-CoV-2 Vaccine. *New England Journal of Medicine*. <https://doi.org/10.1056/nejmoa2035389>
- [Baerends et al. 2023](#) Omicron Variant-Specific Serological Imprinting Following BA.1 or BA.4/5 Bivalent Vaccination and Previous SARS-CoV-2 Infection: A Cohort Study. *Clinical Infectious Diseases*. <https://doi.org/10.1093/cid/ciad402>
- [Banga Ndzouboukou et al. 2025](#) A meta-analysis on the immunogenicity of prototype, monovalent-adapted and bivalent vaccines against SARS-CoV-2 wildtype, Omicron BA.1 and Omicron BA.4/5 in healthy adults. *Virology*. <https://doi.org/10.1016/j.virol.2025.110509>
- [Benkeser et al. 2023](#) Comparing antibody assays as correlates of protection against COVID-19 in the COVE mRNA-1273 vaccine efficacy trial. *Science Translational Medicine*. <https://doi.org/10.1126/scitranslmed.ade9078>
- [Blakney et al. 2021](#) An Update on Self-Amplifying mRNA Vaccine Development. *Vaccines*. <https://doi.org/10.3390/vaccines9020097>
- [Carpp et al. 2024](#) Neutralizing antibody correlate of protection against severe-critical COVID-19 in the ENSEMBLE single-dose Ad26.COVS vaccine efficacy trial. *Nature Communications*. <https://doi.org/10.1038/s41467-024-53727-y>
- [Castaño et al. 2025](#) Distinct components of mRNA vaccines cooperate to instruct efficient germinal center responses. *Cell*. <https://doi.org/10.1016/j.cell.2025.11.023>
- [Chalkias et al. 2022](#) A Bivalent Omicron-Containing Booster Vaccine against Covid-19. *New England Journal of Medicine*. <https://doi.org/10.1056/nejmoa2208343>
- [Chaudhary et al. 2021](#) mRNA vaccines for infectious diseases: principles, delivery and clinical translation. *Nature Reviews Drug Discovery*. <https://doi.org/10.1038/s41573-021-00283-5>
- [Corbett et al. 2020](#) SARS-CoV-2 mRNA vaccine design enabled by prototype pathogen preparedness. *Nature*. <https://doi.org/10.1038/s41586-020-2622-0>
- [Cullis & Felgner 2024](#) The 60-year evolution of lipid nanoparticles for nucleic acid delivery. *Nature Reviews Drug Discovery*. <https://doi.org/10.1038/s41573-024-00977-6>
- [Faksova et al. 2024](#) COVID-19 vaccines and adverse events of special interest: A multinational Global Vaccine Data Network (GVDN) cohort study of 99 million vaccinated individuals. *Vaccine*. <https://doi.org/10.1016/j.vaccine.2024.01.100>
- [Feikin et al. 2022](#) Duration of effectiveness of vaccines against SARS-CoV-2 infection and COVID-19 disease: results of a systematic review and meta-regression. *The Lancet*. [https://doi.org/10.1016/s0140-6736\(22\)00152-0](https://doi.org/10.1016/s0140-6736(22)00152-0)
- [Fong et al. 2023](#) Immune correlates analysis of the PREVENT-19 COVID-19 vaccine efficacy clinical trial. *Nature Communications*. <https://doi.org/10.1038/s41467-022-35768-3>
- [Gilbert et al. 2022](#) Immune correlates analysis of the mRNA-1273 COVID-19 vaccine efficacy clinical trial. *Science*. <https://doi.org/10.1126/science.abm3425>

- Goryaynov & Gurova 2024** Effect of Platform Type on Clinical Efficacy of SARS-CoV-2 Vaccines in Prime Vaccination Settings: A Systematic Review and Meta-Regression of Randomized Controlled Trials. *Vaccines*. <https://doi.org/10.3390/vaccines12020130>
- Guo et al. 2023** Incidence of myopericarditis after mRNA COVID-19 vaccination: A meta-analysis with focus on adolescents aged 12–17 years. *Vaccine*. <https://doi.org/10.1016/j.vaccine.2023.05.049>
- Haensler et al. 2025** Not so cold! Improving the thermostability of mRNA vaccines. *Expert Review of Vaccines*. <https://doi.org/10.1080/14760584.2025.2596674>
- Hertoghs et al. 2025** Vaccine-induced T cell responses correlate with reduced risk of severe COVID-19 in a placebo-controlled efficacy trial. *eBioMedicine*. <https://doi.org/10.1016/j.ebiom.2025.105809>
- Ho et al. 2024** A randomized trial comparing safety, immunogenicity and efficacy of self-amplifying mRNA and adenovirus-vector COVID-19 vaccines. *npj Vaccines*. <https://doi.org/10.1038/s41541-024-01017-5>
- Hou et al. 2021** Lipid nanoparticles for mRNA delivery. *Nature Reviews Materials*. <https://doi.org/10.1038/s41578-021-00358-0>
- Hwang et al. 2025** Limitations of neutralizing antibody titers in COVID-19 vaccine efficacy trials and a call for additional correlates of protection. *Human Vaccines & Immunotherapeutics*. <https://doi.org/10.1080/21645515.2025.2473795>
- Hô et al. 2024** Safety, immunogenicity and efficacy of the self-amplifying mRNA ARCT-154 COVID-19 vaccine: pooled phase 1, 2, 3a and 3b randomized, controlled trials. *Nature Communications*. <https://doi.org/10.1038/s41467-024-47905-1>
- Ip et al. 2024** Cohort study of cardiovascular safety of different COVID-19 vaccination doses among 46 million adults in England. *Nature Communications*. <https://doi.org/10.1038/s41467-024-49634-x>
- Jin et al. 2025** mRNA vaccine sequence and structure design and optimization: Advances and challenges. *Journal of Biological Chemistry*. <https://doi.org/10.1016/j.jbc.2024.108015>
- Jo et al. 2026** mRNA vaccines engage unconventional pathways in CD8+ T cell priming. *Nature*. <https://doi.org/10.1038/s41586-026-10353-6>
- Karikó et al. 2005** Suppression of RNA Recognition by Toll-like Receptors: The Impact of Nucleoside Modification and the Evolutionary Origin of RNA. *Immunity*. <https://doi.org/10.1016/j.immuni.2005.06.008>
- Karikó et al. 2011** Generating the optimal mRNA for therapy: HPLC purification eliminates immune activation and improves translation of nucleoside-modified, protein-encoding mRNA. *Nucleic Acids Research*. <https://doi.org/10.1093/nar/gkr695>
- Kim et al. 2022** Germinal centre-driven maturation of B cell response to mRNA vaccination. *Nature*. <https://doi.org/10.1038/s41586-022-04527-1>
- Klein et al. 2021** Surveillance for Adverse Events After COVID-19 mRNA Vaccination. *JAMA*. <https://doi.org/10.1001/jama.2021.15072>
- Kurhade et al. 2023** Low neutralization of SARS-CoV-2 Omicron BA.2.75.2, BQ.1.1 and XBB.1 by parental mRNA vaccine or a BA.5 bivalent booster. *Nature Medicine*. <https://doi.org/10.1038/s41591-022-02162-x>
- Lamoot et al. 2023** Successful batch and continuous lyophilization of mRNA LNP formulations depend on cryoprotectants and ionizable lipids. *Biomaterials Science*. <https://doi.org/10.1039/d2bm02031a>
- Lederer et al. 2020** SARS-CoV-2 mRNA Vaccines Foster Potent Antigen-Specific Germinal Center Responses Associated with Neutralizing Antibody Generation. *Immunity*. <https://doi.org/10.1016/j.immuni.2020.11.009>
- Leroux-Roels et al. 2026** Efficacy and Safety of an mRNA Seasonal Influenza Vaccine in Adults. *New England Journal of Medicine*. <https://doi.org/10.1056/nejmoa2516491>
- Liu et al. 2021** Safety and immunogenicity of heterologous versus homologous prime-boost schedules with an adenoviral vectored and mRNA COVID-19 vaccine (Com-COV): a single-blind, randomised, non-inferiority trial. *The Lancet*. [https://doi.org/10.1016/s0140-6736\(21\)01694-9](https://doi.org/10.1016/s0140-6736(21)01694-9)
- Ma et al. 2025** Immune correlates analysis of mRNA-1345 RSV vaccine efficacy clinical trial. *Nature Communications*. <https://doi.org/10.1038/s41467-025-61153-x>
- Mudd et al. 2022** SARS-CoV-2 mRNA vaccination elicits a robust and persistent T follicular helper cell response in humans. *Cell*. <https://doi.org/10.1016/j.cell.2021.12.026>
- Mulroney et al. 2024** N1-methylpseudouridylation of mRNA causes +1 ribosomal frameshifting. *Nature*. <https://doi.org/10.1038/s41586-023-06800-3>
- Muro et al. 2026** A distinct antigen presentation pathway drives potent T cell immunity in lipid nanoparticle-based mRNA vaccines. *Science Advances*. <https://doi.org/10.1126/sciadv.aec7827>
- Oster et al. 2022** Myocarditis Cases Reported After mRNA-Based COVID-19 Vaccination in the US From December 2020 to August 2021. *JAMA*. <https://doi.org/10.1001/jama.2021.24110>
- Pardi & Krammer 2024** mRNA vaccines for infectious diseases — advances, challenges and opportunities. *Nature Reviews Drug Discovery*. <https://doi.org/10.1038/s41573-024-01042-y>
- Patone et al. 2022** Risks of myocarditis, pericarditis, and cardiac arrhythmias associated with COVID-19 vaccination or SARS-CoV-2 infection. *Nature Medicine*. <https://doi.org/10.1038/s41591-021-01630-0>
- Polack et al. 2020** Safety and Efficacy of the BNT162b2 mRNA Covid-19 Vaccine. *New England Journal of Medicine*. <https://doi.org/10.1056/nejmoa2034577>
- Risma et al. 2021** Potential mechanisms of anaphylaxis to COVID-19 mRNA vaccines. *Journal of Allergy and Clinical Immunology*. <https://doi.org/10.1016/j.jaci.2021.04.002>
- Rosa et al. 2021** mRNA vaccines manufacturing: Challenges and bottlenecks. *Vaccine*. <https://doi.org/10.1016/j.vaccine.2021.03.038>

- Rozman et al. 2026** N1-Methylpseudouridine directly modulates translation dynamics. *Nature*. <https://doi.org/10.1038/s41586-025-09945-5>
- Sahin et al. 2020** COVID-19 vaccine BNT162b1 elicits human antibody and TH1 T cell responses. *Nature*. <https://doi.org/10.1038/s41586-020-2814-7>
- Selva et al. 2026** Elevated SARS-CoV-2 IgG4 in plasma and mucosa following repeated mRNA boosters impact antibody functions to Omicron and sarbecoviruses. *eBioMedicine*. <https://doi.org/10.1016/j.ebiom.2025.106087>
- Song et al. 2024** Finite immune imprinting on neutralizing antibody responses to Omicron subvariants by repeated vaccinations. *International Journal of Infectious Diseases*. <https://doi.org/10.1016/j.ijid.2024.107198>
- Stuart et al. 2022** Immunogenicity, safety, and reactogenicity of heterologous COVID-19 primary vaccination incorporating mRNA, viral-vector, and protein-adjuvant vaccines in the UK (Com-COV2): a single-blind, randomised, phase 2, non-inferiority trial. *The Lancet*. [https://doi.org/10.1016/s0140-6736\(21\)02718-5](https://doi.org/10.1016/s0140-6736(21)02718-5)
- Tang et al. 2022** Respiratory mucosal immunity against SARS-CoV-2 after mRNA vaccination. *Science Immunology*. <https://doi.org/10.1126/sciimmunol.add4853>
- Turner et al. 2021** SARS-CoV-2 mRNA vaccines induce persistent human germinal centre responses. *Nature*. <https://doi.org/10.1038/s41586-021-03738-2>
- Whitaker et al. 2023** mRNA vaccines against respiratory viruses. *Current Opinion in Infectious Diseases*. <https://doi.org/10.1097/qco.0000000000000948>
- Zhang et al. 2024** Omicron COVID-19 immune correlates analysis of a third dose of mRNA-1273 in the COVE trial. *Nature Communications*. <https://doi.org/10.1038/s41467-024-52348-9>

Every citation in this review was resolved and verified through Crossref, including retraction screening. Generated with the scientific-review skill.